Supplementary Material for: Optimising Binary HVAC Switching in Saudi Residential Buildings with Model Predictive Control and Rule-Constrained PPO

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Equation, section and table numbers without the prefix "S" refer to the main text. Every saving in this supplement is generated from the results files under the evaluation protocol of the main text (Rules A–C), with the code version recorded in each file.

S1 Thermal model: validation, capacity and actuation interval

Validation literature. Bacher and Madsen (Energy and Buildings 43, 2011, 1511–1522) identified grey-box RC models of a measured building by forward selection with likelihood-ratio tests; they rejected the single-state model, whose residuals were not white, and selected a four-state model. The single node used here follows their control-oriented framework but not their selected order. Shi et al. (2022) used an RC network for multi-zone pre-cooling optimisation, and Wang et al. (2021) validated a second-order thermal network for residential buildings (both in the main text's reference list). The main text's results are therefore reported with the capacity sweeps of Table S1(b) and of the main text's sensitivity table, not on the strength of a validated single node.

Capacity. At an hourly interval, capacities of 77.2 (air mass), 500 and 1,200 kJ/K give per-interval excursions of 93.3, 14.4 and 6.00 °C and violate the main text's actuation condition, so a sweep there would compare configurations that admit no comfort-feasible schedule. Table S1(b) spans the capacities around the reference at $\Delta t = 15$ min and marks each by the two-sided condition (8) of the main text: its lower limit needs $C_{\mathrm{th}} \geq Q_{\mathrm{hvac}} \Delta t / \delta_{\mathrm{db}} = 1{,}800$ kJ/K, and its upper limit, with the one-interval warming excursion at the 690 W peak load, is 0.99 °C at 2,400 kJ/K, so THERM's 1 °C deadband fails it narrowly there. THERM's comfort record is reported in every row so that an infeasible configuration cannot be mistaken for a cheap one. The headline 4,800 kJ/K is 50 kJ m^{-2}K^{-1} over 96 m^2 of participating surface, which gives a time constant of 56 h and a free-float time across the 2 K band of 4.3 h at the ninety-fifth percentile of seasonal load and 8.0 h at the median. ISO 13786's areal capacity is a 24 h quantity; the capacity effective at the cycle period is lower.

Actuation interval. With the exact update the interval carries no integration error; it decides how often a controller may act and whether THERM can hold its own deadband. THERM begins to leak at 30 min and fails at 1 h, where the two arms are no longer held to the same constraint. The saving is *not* independent of the interval. A finer interval lets the tuned thermostat widen its deadband (to 1.78–1.79 °C at 5 min, against 0.94–0.99 °C at 30 min) while THERM keeps its fixed one (Table S1(a)). TUNED saves about 25% at 5 min and −1.2 to −1.5% at 30 min (main text, Section 9.3 and Table 8), where THERM's deadband fails the upper limit of (8) narrowly (0.99 °C) and THERM leaks. Comparisons with studies at a different interval must take that into account.

Stability and the discrete tariff. Under the exact update the one-step map is $e^{\mathbf{M}\Delta t}$, with spectral radius $\exp(-\Delta t\, \mu_{\min}/C_{\mathrm{th}}) < 1$ for every $\Delta t > 0$, where $\mu_{\min}$ is the smallest eigenvalue of $\operatorname{diag}\Lambda + L_{\mathrm{iz}}$. Forward Euler has map $I + \Delta t \mathbf{M}$ and requires $\Delta t < 2C_{\mathrm{th}}/\mu_{\max}$. At $\Delta t = 1$ h, with

Table S1: Admissibility sweeps (Riyadh, tariff in force). "adm.": THERM's 1 °C deadband against both limits of condition (8) of the main text. THERM's occupied degree-hours are a property of the baseline alone, from a separate THERM run on the first held-out Riyadh week (2–9 May). Bold marks the adopted configuration. Savings (%) under the main text's protocol, every arm refitted; B^* under V_N, rounded up; Jeddah and Taif are in the main text's actuation-step table. [b]801 grid nodes, pooled 0.10 K fallback. [c]$\lambda/\lambda^* = 6.27$ (see the main text's actuation-step table).

(a) Actuation interval, $C_{\mathrm{th}} = 4{,}800\,\mathrm{kJ/K}$				
Δt (min)	$Q_{\mathrm{hvac}}\Delta t/C_{\mathrm{th}}$ (°C)	adm.	THERM deg-h	TUNED / MPC-24 h / B^*
5	0.125	yes	0.000	25.0 / 23.8[b] / 27.3
15	**0.375**	**yes**	**0.000**	12.5 / 18.3 / 20.0
30	0.750	no (upper)	0.025	−1.2 / 6.6[c] / 10.9
60	1.500	no	4.470	—
(b) Thermal capacity, $\Delta t = 15\,\mathrm{min}$				
C_{th} (kJ/K)	$Q_{\mathrm{hvac}}\Delta t/C_{\mathrm{th}}$ (°C)	adm.	THERM deg-h	TUNED / MPC-24 h / B^*
1,200	1.500	no	2.737	—
1,800	1.000	no	2.500	—
2,400	0.750	no (upper)	0.000	−2.8 / 12.2 / —
4,800	**0.375**	**yes**	**0.000**	12.5 / 18.3 / 20.0
7,200	0.250	yes	0.000	16.0 / 19.8 / 20.0
9,600	0.188	yes	0.000	15.8 / 16.2 / —

the conductances of the main text's Table 2 and $\lambda_{ij} = 12.5\,\mathrm{W/K}$, that is $C_{\mathrm{th}} > 40.8$–$44.3\,\mathrm{kJ/K}$ for the single zone and $C_{\mathrm{th}} > 85.8$–$185.1\,\mathrm{kJ/K}$ for the coupled topologies (1×2 to 4×4, three cities); the air-only capacity of 77.2 kJ/K meets it for the single zone and violates it for every coupled topology. The scheduler acts on a grid, so the realised bill is $\sum_t p(E^{(t)}_{\mathrm{cum}})\Delta E^{(t)}$, which equals the integral of the main text's path proposition only when no tier boundary falls strictly inside an interval's increment; where one does, the discrete bill is path-dependent at the order of one interval's consumption (0.17 kWh here, against tier widths of 2,000 kWh). For the account modelled no boundary is approached, so the discrete and continuous statements coincide.

S2 The SLHA abstraction and the corollary's mean

An SLHA requires a constant flow in each mode, whereas the rates of the main text's SLHA definition depend on T through $\Lambda(T_{\mathrm{out}} - T)$. Reading them as constants evaluates them at a fixed reference temperature. With $\Lambda = 23.75\,\mathrm{W/K}$ and $C_{\mathrm{th}} = 4{,}800\,\mathrm{kJ/K}$ ($\tau = 56.1\,\mathrm{h}$) the exact free-floating rise time $\tau \ln[(T_{\mathrm{out}} - T_{\mathrm{min}})/(T_{\mathrm{out}} - T_{\mathrm{max}})]$ across the band is 9.378, 6.612 and 5.107 h at outdoor temperatures of 35, 40 and 45 °C, against 9.356, 6.604 and 5.103 h from the constant-rate expression, errors of 0.2%, 0.1% and 0.1%. A wider band, a smaller capacity or a leakier envelope would widen the gap.

Under constant drift each linear phase of a cycle averages to the midpoint of its band, so the maximal-width cycle attains the mean $(T^g_{\mathrm{on}} + T^g_{\mathrm{off}})/2$ and no other. On a Riyadh held-out week's mean disturbance with guards at 23.60/22.40 °C, exponential rather than linear drift moves the cycle mean 0.006 K from the midpoint. The 15-min step matters more: a switch occurs at the first sample past a threshold, so the zone overshoots the off threshold by up to one interval's cooling (0.29 K here) and the sampled cycle mean lies 0.09 K below the midpoint. Matching is therefore done on the simulated trajectory, never on the midpoint formula.

Table S2: Protocol and device sensitivity, pooled equalised saving per city. ^{b}fitted at the pooled 0.10 K fallback. The receding-horizon controllers were not refitted under the per-window criterion.

Variant	Riyadh / Jeddah / Taif
TUNED refitted under a per-window 0.03 K criterion	9.7 / 12.2 / 4.0
Guard refitted under a per-window 0.03 K criterion	5.8 / 5.9 / 7.8^{b}
F4 (guard − TUNED) under that criterion	−4.0 / −6.3 / +3.8
TUNED on a 0.1 °C hardware grid (all at 23.6/22.4)	5.7 / 7.4 / 8.4
Rule C, cheapest vs dearest crossing: TUNED	12.5–13.7 / 13.6–14.4 / 12.4 (both)
Rule C, cheapest vs dearest crossing: MPC-24 h (perfect)	18.3–19.0 / 18.1–18.4 / 16.9–17.4
THERM scored against its own curve	−0.8 / −0.1 / 0.0
MPC-6 h at $\lambda^* \cdot \{0.99, 1.01, 1.05\}$ (s; $\Delta\tilde{T}$ in K)	$16.2, 14.0, 10.7$ / $17.4, 16.5, 10.5$ / $15.0, 14.0, 7.7$ ($+0.198, -0.026, -0.126$ / $+0.185, +0.018, -0.109$ / $+0.293, -0.059, -0.184$)
MPC-24 h at $\lambda^* \cdot \{0.99, 1.01, 1.05\}$ (s; $\Delta\tilde{T}$ in K)	$18.4, 18.0, 16.9$ / $18.2, 17.9, 16.0$ / $14.4, 17.5, 16.8$ ($+0.185, +0.015, -0.043$ / $+0.168, +0.033, -0.056$ / $+0.379, +0.018, -0.042$)
MPC-24 h at 1,601 / 3,201 grid nodes, fitted λ	18.3/18.2 / 18.1/18.5 / 16.9/16.6
Back-off cells† (b, K): MPC 0.25/1/2 h	—, 0.21, — / 0.205, 0.25, — / —, 0.345, 0.345
D1 latch, margin $\sigma = 0/0.05/0.1$ K	$14.0, 12.4, 6.2$ / $13.5, 8.3, 7.7$ / $12.4, 10.5, 7.0$
MPC 1/6/24 h at the guard's fitted rules	$2.5, 6.4, 6.7$ / $1.5^{b}, 7.6, 4.4^{b}$ / $8.0^{b}, 9.5^{b}, 10.5$

S3 Protocol sensitivity

S4 Clairvoyant match per window

The clairvoyant multiplier is bisected per held-out window towards THERM's clipped mean; at the critical multiplier the schedule's mean jumps by 0.1–0.3 K, so where the 0.002 K target cannot be met the closest schedule no warmer than THERM is kept.

Table S3: CLV match per held-out window (zero-based day indices 121–289, starting 2 May): `tol_met` (T/F) and residual $\tilde{T}_{\mathrm{CLV}} - \tilde{T}_{\mathrm{THERM}}$ (K), under the V_N terminal. The 0.002 K target is met in 3 of the 21 windows (2 in Riyadh, 1 in Jeddah, none in Taif).

City	Windows 1–7: `tol_met` / residual
Riyadh	F/−0.013, F/−0.029, T/−0.001, T/+0.000, F/−0.026, F/−0.034, F/−0.148
Jeddah	F/−0.010, F/−0.076, F/−0.025, T/−0.001, F/−0.003, F/−0.011, F/−0.003
Taif	F/−0.018, F/−0.019, F/−0.002, F/−0.007, F/−0.075, F/−0.028, F/−0.025

S5 Orientation and actual weather years

The headline zone's glazed face is south-facing. The full protocol is repeated in Riyadh with the glazed face turned to each cardinal direction, envelope, plant and windows unchanged: TUNED saves 13.3, 12.7, 12.5 and 12.2% facing north, east, south and west, all fitted at pooled 0.03 K.

A typical year is not any year that occurred, so fifteen actual years per city (2009–2023, ERA5 reanalysis on a 0.25° cell, shifted by one hour to align its timestamps with the TMYx convention, with the same plane-of-array transposition) are used to measure spread at fixed method and plant. From these the mildest, median and hottest years by cooling degree-days at base 24 °C are run in two ways (ERA5 is a reanalysis, not the station record behind TMYx, and some of these years fall within the TMYx period of record, so the spread mixes year-to-year variation with dataset differences): *transfer* applies the TMYx-tuned parameters unchanged, as a thermostat commissioned once would

experience, and *re-tuned* repeats Rule A inside the year. TUNED saves (transfer / re-tuned, %; years in the order mildest, median, hottest): Riyadh 2013 12.4/14.4, 2019 12.0/13.8, 2021 12.8/11.5; Jeddah 2011 13.2/14.5, 2015 13.4/12.3, 2023 13.4/14.8; Taif 2013 10.5/14.3, 2018 12.0/13.7, 2017 11.9/14.4, all at pooled 0.03 K and with zero over-temperature. In Taif's mild years 2013 and 2018 the zone free-floats below the band for 6.7–6.9 and 2.6–3.0 °C·h. The TMYx figures remain the reported ones.

S6 Latent load

A two-state zone model (sensible temperature and moisture, with hygric buffering and a sensible-heat-ratio-limited coil) was built on the headline parameters to test the sensible-only idealisation. Its coil did not reduce sensible capacity when removing moisture, it placed the whole dwelling's infiltration moisture in one room, and its tuning did not follow the protocol of the main text. No result from it is reported, and the latent axis is treated as untested. A coil with a capacity map and a per-room moisture balance would be needed before any latent comparison.

S7 Multi-zone dwellings

Riyadh, independent per-room units, the protocol of the main text applied to the dwelling's mean occupied temperature, $\lambda_{ij} = 12.5$ W/K with 35 W/K as a sensitivity. MPC-MZ is a non-cooperative distributed receding-horizon controller with one-step-delayed plan exchange: each room solves its own single-zone programme against its neighbours' previously announced plans and minimises its own cost. It has no optimality guarantee, and no bound exists for these layouts.

Table S-MZ: Multi-zone results, Riyadh, independent per-room units: pooled equalised saving on the dwelling mean (%). THERM is scored against its own comfort-equalised curve (the conservatism of Rule C). MPC-MZ: 24 h look-ahead, perfect forecast, 1601 nodes, one comfort weight shared by the rooms. [a]pooled 0.03 K; [b]pooled 0.10 K (fallback). TUNED's mean spread of room temperatures is 0.03–0.08 K in every layout.

Layout	Glazed faces	THERM / TUNED / guard / MPC-MZ
1×1	S	-0.8 / 12.5^a / 5.7^a / 18.3^a
1×2, $\lambda_{ij} = 12.5$	S, W	-0.1 / 12.4^a / 7.2^a / 16.0^b
1×2, $\lambda_{ij} = 35$	S, W	-0.3 / 12.2^a / 6.5^a / 15.8^b
2×2, $\lambda_{ij} = 12.5$	S+E, S+W, N+E, N+W	-0.1 / 11.7^a / 8.3^a / 15.9^a
2×2, $\lambda_{ij} = 35$	S+E, S+W, N+E, N+W	-0.2 / 12.9^a / 7.6^a / 16.1^a
1×4, $\lambda_{ij} = 12.5$	S, S, N, N	-0.5 / 13.6^a / 7.7^a / 15.7^a
1×4, $\lambda_{ij} = 35$	S, S, N, N	-0.3 / 13.3^a / 6.9^a / 16.1^a

S8 Comfort records and the bound at each arm's own mean

The cold rule is a hard constraint for the optimisers (dynamic programme, receding-horizon controllers, clairvoyant schedule and bound). The rule-layer arms and the D1 shield are designed to meet it from the nominal plant parameters, and the threshold arms meet it on the tuning windows by Rule A. Table S4 lists every non-zero commanded under-temperature on the held-out windows; every other arm and configuration records zero, and every fitted arm records zero occupied over-temperature except the multi-zone controller in the 1×4 row at $\lambda_{ij} = 35$ W/K (0.0001 °C·h), whose rooms plan against their neighbours' plans of the previous step. Counting the vacation weeks as fully occupied lowers the setback savings to 16.1, 17.4 and 14.3% for TUNED-SB and 20.8, 20.5 and 19.5% for MPC-SB (Riyadh, Jeddah, Taif). Table S5 gives the bound B_a at the principal arms' own mean temperatures, with each arm's saving rescored under the V_N terminal; all 89 arm and seed rows satisfy $s_a \leq B_a$.

Table S4: Commanded under-temperature (degree-hours below $T_{\min}$ at the end of cooling steps, °C·h summed over the seven held-out windows), every non-zero value. Headline: $\Delta t = 15\,\text{min}$, $C_{\text{th}} = 4{,}800\,\text{kJ/K}$, 1 °C comparator. TUNED's value is unchanged across the switching-charge sweep.

Configuration, arm	Riyadh / Jeddah / Taif
Headline, TUNED	0 / 0.0007 / 0.0066
Headline, TUNED-SB (setback)	0.0102 / 0 / 0
Headline, MPC 6 h persistence	0.0006 / — / 0
Headline, all other arms of Tables 3 and 4	0 / 0 / 0
TUNED at $\Delta t = 5$ / 30 min	0, 0.0097 / 0, 0.0025 / 0.0011, 0
TUNED at $C_{\text{th}} = 2{,}400$ / 7,200 kJ/K	0, 0.0047 / 0, 0 / 0.0042, 0.0017
TUNED against a 0.5 / 0.75 / 1.25 °C comparator	0.0054, 0, 0.0092 / 0.0063, 0.0016, 0.0063 / 0.0066, 0.0066, 0
TUNED, per-window refit; north façade	0; 0.0026 / 0.0016; — / 0; —
TUNED, actual years (mildest, median, hottest), transfer	0, 0, 0 / 0.0007, 0, 0.0001 / 0.0110, 0.0034, 0.0015
TUNED, actual years, re-tuned	0, 0.0007, 0.0034 / 0.0023, 0, 0.0238 / 0, 0, 0
TUNED, 2×2 dwelling, $\lambda_{ij} = 12.5$ / 35 W/K	0.0002 / 0.0004 (Riyadh only)
MPC-MZ, 2×2 dwelling, $\lambda_{ij} = 12.5$ / 35 W/K	0.000003 / 0.000025 (Riyadh only)
PPO without rules, the one admissible seed	0.0065 / — / —

Table S5: Saving s_a rescored under the V_N terminal and the bound B_a at each arm's own pooled clipped mean (%), held-out windows, headline configuration. End-of-window temperatures (°C, pooled): CLV 22.33, 22.40, 22.35; THERM 22.92, 22.77, 23.03; TUNED 22.83, 23.05, 22.69; MPC 24 h (perfect) 22.87, 23.18, 23.12 in Riyadh, Jeddah and Taif.

Arm	s_a / B_a: Riyadh; Jeddah; Taif
TUNED	12.7/19.7; 13.2/20.2; 12.9/20.8
Guard hysteresis	6.0/19.6; 7.6/20.2; 11.0/21.4
D1 shield on latch ($\sigma = 0$)	13.4/19.7; 14.0/20.1; 12.5/20.6
MPC 24 h, perfect	18.6/19.9; 17.2/19.3; 16.3/20.4
MPC 24 h, persistence	13.0/18.1; 14.1/18.9; 11.8/19.9
CLV (Rule C)	18.3/19.6; 18.5/19.4; 16.5/20.2

S9 Direction of bias of each idealisation, and the ablation

Table S6: Direction in which each idealisation moves the tuned thermostat's reported saving, with the evidence in this study.

Idealisation	Direction	Evidence
Comparator deadband assumed 1 °C	overstates if wider	5.7% against 1.25 °C (main Table 7(c))
Charge 0.15 SAR per on-command	overstates	3.8–11.0% at 0.025–0.114 SAR (Table 7(b))
Switch points on a 0.01 °C grid	overstates	5.7–8.4% on a 0.1 °C grid (Table S2)
Reference capacity (24 h ISO value)	probably overstates	−2.8% at 2,400 kJ/K (Table 7(d))
Actuation step 15 min	either way	about 25% at 5 min, −1.2 to −1.5% at 30 min (Table 8)
Admission tolerance pooled 0.03 K	either way	9.7, 12.2, 4.0% under a per-window criterion (Table S2)
Rule C cheapest crossing	understates	THERM self-scores −0.8, −0.1, 0.0% (Table S2)
No start-up energy	understates energy saving	0.33–0.90% for 10–25 Wh per start (main Section 10.4)
Noise-free sensing	favours fine thresholds	not quantified
Perfect model and forecast	favours MPC	persistence margin not resolved (main Table 5, F3)
Single node; constant capacity and COP; no internal gains; sensible only	not established	not swept
One typical year	—	actual years: 10.5–13.4% transfer, 11.5–14.8% retuned (S5)

Table S7: Ablation of the rule layer, the learner, the one-step model and the horizon (pooled equalised saving, %, from the main text's Table 3 and Section 9.2). [b]fitted at the pooled 0.10 K fallback.

Configuration	Riyadh / Jeddah / Taif
PPO without rules (admissible seeds)	−3.5 (1 of 5) / none of 5 / none of 5
PPO inside the training rules (5 seeds)	5.7 / 7.6 / 8.6–11.0
Guard hysteresis (rules, no learner)	5.7 / 7.6 / 11.0
D1 shield on latch (one-step model, no rules)	14.0 / 13.5 / 12.4
D1 shield on the trained policies (5 seeds)	12.8–14.1 / 11.2–13.5 / 8.0–12.0
MPC 24 h inside the guard's fitted rules	6.7 / 4.4[b] / 10.5
MPC 24 h inside the training rules	11.6[b] / 14.1 / 10.0[b]
TUNED (no rules, no model)	12.5 / 13.6 / 12.4
E2 (comfort term and forecast; Riyadh, 3 seeds)	not comfort-matched; not tabulated